

IPSITA BOSE

Perception Engineer | Multimedia Designer

PROFILE

Perception Engineer and Multimedia Designer with an Electronics Engineering background and multidisciplinary experience across machine vision, industrial automation, engineering simulation, interactive 3D web development, UX/UI design, graphic design, motion, video production, technical storytelling, content strategy, and voice performance. Works at the intersection of sensing, computation, simulation, visualization, communication, and human perception— translating physical systems, machine states, engineering principles, software workflows, and abstract information into experiences that people can perceive, understand, navigate, and act on. Hands-on experience spans industrial machine-vision systems, OCR-based inspection, camera and software integration, OPC-connected automation environments, MES and SCADA workflows, MATLAB-based thermodynamic simulations, browser-based 3D applications, digital-twin visualization, interface design, frontend implementation, process animation, visual communication, branded content, commercial graphic design, technical video production, and studio voice-over. Combines engineering reasoning with design, interaction, storytelling, and multimedia production to bridge machines, software, information, brands, and human users.

CORE CAPABILITIES

Perception Engineering: Machine Vision, Industrial Vision Systems, Human-Machine Perception, Visual Inspection Workflows, OCR-Based Inspection, Image-Based Inspection, Machine-Generated Visual Feedback, Perception-Oriented Interface Design, Information Visualization, Human Interpretation of Machine States, Visual Encoding of Process Conditions, Anomaly Visualization, Human-Machine Interfaces, Operator Interfaces, Interaction Feedback, System-State Communication, Perceptual Hierarchy, Visual Attention, Information Prioritization, Visual Feedback Systems, Sensor-to-Interface Workflows, Machine-to-Human Communication, Physical-to-Digital Representation

Machine Vision Systems: Cognex In-Sight Vision Systems, Cognex In-Sight 7402, Cognex IS2000, Industrial Cameras, Camera Communication, OCR Configuration, Character Recognition Workflows, Visual Inspection Logic, Pass/Fail Inspection Logic, CountPassFail Mapping, Inspection Parameter Configuration, Label Verification, Packaging Inspection, Machine Vision Integration, Vision Job Configuration, Camera-to-Software Communication, OPC-Based Vision Data Exchange, Inspection Result Mapping, Operator-Facing Inspection Feedback, Vision System Troubleshooting, Industrial Quality Inspection

Electronics Engineering: Electronics Engineering, Electronic Systems Fundamentals, Sensors and Instrumentation Awareness, Digital Electronics, Analog Electronics Awareness, Signals and Systems Awareness, Embedded Systems Awareness, Measurement Systems, Industrial Instrumentation Awareness, Electrical and Electronic Interfaces, Control-System Fundamentals, Hardware-Software Interaction, System-Level Thinking, Engineering Problem Solving, Technical Troubleshooting, Engineering Documentation, Physical-System Interpretation, Electronic System Integration Awareness

Engineering Simulation and Mathematical Modeling: MATLAB, Engineering Computation, Mathematical Modeling, Thermodynamic Simulation, Carnot Cycle Analysis, Quasistatic Process Modeling, Thermodynamic State Analysis, Heat and Work Calculations, Idealized Process Simulation, Numerical Problem Solving, Physics-Based Modeling, Equation-Based Simulation, Engineering Visualization, Parameter-Based Analysis, Computational Engineering, Analytical Problem Solving, Technical Model Development

Thermodynamics: Thermodynamic Systems, Carnot Engine, Carnot Cycle, Quasistatic Processes, Reversible Process Concepts, Heat Engines, Thermal Efficiency, State Variables, Pressure-Volume Relationships, Temperature-Entropy Concepts, Work and Heat Transfer, Thermodynamic Process Analysis, Engineering Problem Solving, Mathematical Representation of Physical Processes

Industrial Automation: Industrial Automation, PLC Ecosystems, Automation Workflows, Process Automation, Industrial Software, Control-System Awareness, Industrial Communication, OPC, OPC DA, Machine-to-Software Integration, Production Systems, Manufacturing Operations, Process Visualization, Automation Demonstrations, Industrial Demo Environments, Manufacturing Workflow Analysis, Equipment-State Visualization, Operational Technology Awareness, Industrial Digitalization, Smart Manufacturing, Industry 4.0 Concepts

Industrial Software Platforms: AVEVA System Platform, AVEVA OMI, AVEVA InTouch Awareness, AVEVA IDE Awareness, EcoStruxure, ProLeiT, Manufacturing Execution Systems, MES, SCADA, Human-Machine Interfaces, HMI, Industrial Dashboards, Process Monitoring Interfaces, Production Management Workflows, Batch Manufacturing Workflows, Reporting Interfaces, Simulation Environments, Industrial Visualization Systems, Enterprise Manufacturing Software, Digital Operations Interfaces

Manufacturing Domain Knowledge: Food and Beverage Manufacturing, Dairy Manufacturing, Spray Drying, Batch Processing, Production Lines, Process Cells, Production Orders, Shifts, Line Visualization, Production Dashboards, Downtime Reporting, Timeline Analysis, Measured Value Analysis, Process Optimization Concepts, Label Assurance, Quality Inspection, Life Sciences Manufacturing, Pharmaceutical Manufacturing Awareness, Electronic Batch Manufacturing Records, eBMR, ISA-88 Batch Process Awareness, Electronic Signatures, Audit Trails, Deviation Management, Review by Exception, Production Review, QA Review, Batch Closure

Digital Twins and Process Visualization: Digital Twin Concepts, Industrial Digital Twins, Process Visualization, Equipment Visualization, Operational-State Visualization, Material Flow Visualization, Process Condition Representation, Abnormal-State Visualization, Manufacturing Simulation Visuals, 3D Industrial Demonstrations, Animated Process Scenarios, Interactive Process Communication, KPI-Driven Digital Twin Interfaces, Physical-to-Digital Representation, Digital Representation of Equipment Behaviour, Animated Industrial Workflows, Real-Time System Visualization Concepts

3D Web Development: Three.js, React, JavaScript, HTML, CSS, WebGL Concepts, Browser-Based 3D Rendering, Interactive 3D

Applications, GLTF Workflows, 3D Model Integration, Scene Setup, Camera Systems, Orbit Controls, Spatial Interaction, Interactive Markers, Event-Driven Interaction, Browser-Based Product Experiences, Browser-Based Industrial Experiences, Responsive 3D Interfaces, Frontend State Handling, User Interaction Logic, Interactive Navigation Systems, Web-Based Visualization, GitHub, GitHub Pages, Vite

Frontend Development: HTML, CSS, JavaScript, React, Component-Based Interfaces, Browser-Based Applications, Interactive Interfaces, Responsive UI, Frontend Implementation, UI Rendering, Event Handling, Navigation Logic, State-Driven Interaction, Interface Prototyping, Frontend Debugging, Browser Developer Tools, GitHub Version Control Workflows, Vite, Static Web Deployment, Multimedia Integration, Interface Behaviour Implementation, Interactive Technical Prototypes, Playwright Awareness

Python and Technical Automation: Python, YAML-Driven Content Workflows, Automated Document Generation, Template-Based Resume Generation, File Processing, Structured Data Handling, Script-Based Workflow Automation, LaTeX Generation Workflows, XeLaTeX Automation, Regular Expressions, Data Transformation, Programmatic Content Rendering, Utility Scripts, Technical Workflow Automation

UX Design: UX Design, User-Centered Design, Interaction Design, User Flows, Workflow Design, Information Architecture, Task-Oriented Interface Design, Product Workflow Translation, User Journey Awareness, Interaction Models, Interface Behaviour Definition, User Advocacy, Workflow Evaluation, Usability-Oriented Design, Error-State Consideration, Feedback-State Design, Accessibility Awareness, Human Factors Awareness, Enterprise UX, Industrial UX, Complex-System UX, Technical Product UX

UI Design: UI Design, High-Fidelity Mockups, Low-Fidelity Wireframes, Visual Hierarchy, Interface Layout, Dashboard Design, Data-Dense Interface Design, Product Interface Design, Operator Interface Design, Design Specifications, Implementation-Ready Visual Assets, Component Consistency, Shared Components Awareness, Design Systems Awareness, Design Tokens Awareness, UI Handoff, Design Documentation, Figma, Interactive Prototypes

Dashboard and Information Design: KPI Dashboard Design, Operational Dashboards, Manufacturing Dashboards, Process Dashboards, Production KPI Visualization, Quality KPI Visualization, Efficiency Visualization, Equipment Status Visualization, System Health Visualization, Process Parameter Visualization, Data Hierarchy, Visual Encoding, Status Indicators, State Differentiation, Information Density Management, Operator-Focused Dashboards, Technical Data Communication

Human Factors and Usability: Usability Testing, Usability-Oriented Evaluation, Workflow Testing, Multi-User Workflow Evaluation, Human Factors Awareness, User Feedback Translation, User Impact Assessment, Reliability Issue Documentation, Access Dependency Analysis, Workflow Failure Documentation, Performance Issue Documentation, Interface Behaviour Evaluation, Research Awareness, Customer Feedback Awareness, Interaction Evaluation, User Advocacy, Cognitive Load Awareness, Accessibility Awareness

3D Design and Visualization: Blender, 3D Modeling, 3D Visualization, Product Visualization, Industrial Visualization, Scene Composition, Materials, Lighting, Camera Placement, Camera Animation, Object Animation, Process Animation, Equipment Animation, Environment Design, Spatial Storytelling, Cross-Section Visualization, Product Rendering, Technical 3D Communication, Animated Industrial Scenarios, Looping Animation Design, GLTF Export Workflows

Motion Design: Motion Graphics, Animation Planning, Visual Timing, Technical Animation, Product Demonstration Animation, Explainer Animation, Process Animation, Interface Animation Awareness, Storyboard-to-Motion Translation, Seamless Loop Design, Motion-Based Technical Communication, Adobe After Effects, Blender Animation, AI-Assisted Motion Generation

Video Production: Video Editing, Technical Video Production, Product Demo Videos, Industrial Demo Videos, Explainer Videos, Webinar Videos, Promotional Videos, Technical Walkthroughs, Video Storyboarding, Shot Planning, Voice-Over Planning, Script-to-Visual Mapping, B-Roll Planning, Screen Recording Integration, Motion Graphics Integration, Visual Continuity, Narrative Editing, Adobe Premiere Pro, Adobe After Effects, AI-Assisted Video Generation, TTS-Based Production Workflows

Graphic Design: Graphic Design, Visual Design, Poster Design, Layout Design, Typography, Composition, Visual Hierarchy, Digital Graphics, Marketing Graphics, Technical Graphics, Presentation Graphics, Social Media Graphics, Infographics, Campaign Assets, Brand Communication, Client-Facing Creative Work, Digital Communication, Image Editing, Adobe Photoshop, Adobe Illustrator, Adobe Creative Suite, Canva

Vector Design and Illustration: Adobe Illustrator, Vector Graphics, Vector Asset Creation, Pen Tool Workflows, Shape Construction, Typography, Poster Composition, Graphic Layout, Visual Asset Development, Digital Illustration Awareness, Print-Oriented Graphic Design, Client Design Execution

Presentation and Information Communication: Presentation Design, Technical Presentations, Executive-Facing Visual Communication, Product Presentations, Webinar Decks, Internal Communication Assets, Demo Support Presentations, Information Simplification, Complex Concept Visualization, Engineering-to-Business Communication, Technical Storyboarding, Presentation Narrative, Visual Explanation Systems, PowerPoint

Content Strategy: Content Strategy, Creative Strategy, Content Mapping, Audience-Oriented Communication, Product Storytelling, Technical Storytelling, Narrative Development, Message Hierarchy, Cross-Channel Content Planning, Content Structure, Campaign Content, Educational Content, Webinar Content, Product Communication, Technical Communication Strategy, Perception-Oriented Messaging, Information Simplification, Communication Architecture

Scriptwriting and Storytelling: Scriptwriting, Voice-Over Writing, Technical Scriptwriting, Explainer Scripts, Webinar Scripts, Product Demo Narratives, Storyboarding, Shot Breakdown, Narrative Sequencing, Technical Concept Simplification, Feature-to-Benefit Translation, Audience Adaptation, Visual Storytelling, Product Storytelling, Engineering Storytelling

Voice and Audio Communication: Commercial Voice-Over, Studio Voice Recording, Advertising Spots, Brand Voice Work, Script Interpretation, Vocal Delivery, Microphone Performance, Timing and Pacing, Tone Adaptation, Promotional Voice Work, Commercial Read Style, Directed Recording Sessions, Audio Communication, Spoken Brand Communication, Performance Consistency

Brand and Visual Communication: Brand Communication, Visual Identity Awareness, Consistent Visual Systems, Marketing

Communication, Campaign Communication, Digital Brand Assets, Audience Perception, Visual Tone, Message Consistency, Technical Brand Communication, Product Positioning Awareness, Visual Narrative, Perception Engineering for Communication

Product and Engineering Collaboration: Cross-Functional Collaboration, Engineer Collaboration, Automation Engineer Collaboration, Software Team Collaboration, Product Stakeholder Collaboration, Marketing Collaboration, Sales Collaboration, Subject-Matter Expert Collaboration, Requirement Gathering, Requirement Translation, Technical Requirement Interpretation, Software Handoff, Design Handoff, Product Workflow Clarification, Demo Development, Technical Review, Stakeholder Communication, Iterative Feedback, Agile Awareness, SAFe Agile Awareness

Technical Documentation: Technical Documentation, Workflow Documentation, User-Facing Documentation, Demo Documentation, Issue Documentation, Test Findings, Implementation Notes, Interface Specifications, Process Descriptions, Technical Presentations, Engineering Communication, Visual Documentation, Product Walkthrough Documentation, Handover Documentation

Development and Production Tools: MATLAB, Python, Visual Studio Code, GitHub, GitHub Pages, Figma, Blender, Adobe Photoshop, Adobe Illustrator, Adobe Premiere Pro, Adobe After Effects, Adobe Creative Suite, Microsoft Excel, Microsoft PowerPoint, Microsoft 365, SharePoint, Browser Developer Tools, YAML, LaTeX, XeLaTeX, JavaScript, React, Three.js, HTML, CSS, Vite

Emerging and AI-Assisted Workflows: AI-Assisted Video Generation, Generative Visual Workflow Awareness, Prompt-Based Creative Production, AI-Assisted Storyboarding, AI-Assisted Technical Communication, TTS Voice-Over Workflows, Generative Media Integration, AI-Assisted Ideation, Human-in-the-Loop Creative Production

EXPERIENCE

SYSTEM & ARCHITECTURE ENGINEER | Industrial Automation & Digital Technology, India

02/2025 – Present

- Worked across industrial automation, machine vision, enterprise software, UX/UI, frontend development, interactive visualization, 3D production, technical communication, and digital demonstration systems.
- Integrated and configured industrial machine vision workflows involving Cognex cameras, OCR inspection, inspection parameters, pass/fail logic, camera communication, OPC mapping, and operator-facing industrial software.
- Restored and troubleshoot communication between industrial vision hardware and software environments while validating inspection behaviour and downstream data exchange.
- Translated vision-system outputs and machine-generated states into interfaces and visual feedback understandable by operators and demonstration audiences.
- Built browser-based interactive experiences using HTML, CSS, JavaScript, React, Three.js, Vite, GitHub, and modern frontend workflows.
- Developed interactive 3D web applications integrating GLTF models, camera systems, orbit controls, markers, routed navigation, animation states, and user interaction logic.
- Created 3D process, product, equipment, and digital-twin visualizations using Blender for industrial manufacturing and automation use cases.
- Developed animated representations of material flow, equipment behaviour, process states, and abnormal operating conditions for digital-twin demonstrations.
- Designed dashboard concepts, KPI screens, interface layouts, high-fidelity mockups, product flows, and implementation-ready visual assets for industrial and enterprise software.
- Worked with MES, SCADA, HMI, ProLeiT, AVEVA, EcoStruxure, manufacturing, production, and reporting workflows.
- Translated complex industrial workflows into structured interaction models and user-facing digital experiences.
- Conducted usability-oriented evaluation of multi-user enterprise software workflows across virtualized environments.
- Documented workflow failures, reliability issues, access dependencies, application behaviour, performance gaps, and their impact on users.
- Created interface visuals and supporting assets for production dashboards, process monitoring, reporting workflows, dynamic line visualization, downtime analysis, timeline analysis, measured-value analysis, and simulation controls.
- Produced technical product demonstrations combining industrial systems, software interfaces, 3D visualization, video, motion, and storytelling.
- Produced scripts, storyboards, voice-over structures, shot plans, explainer videos, walkthroughs, webinar media, product communication, and promotional content.
- Developed visual narratives for manufacturing, smart manufacturing, digital transformation, process automation, life sciences, food and beverage, and industrial software topics.
- Produced technical graphics, animations, presentation systems, product visuals, and marketing-support assets using Figma, Adobe Creative Suite, Blender, and video-production tools.
- Collaborated with engineers, automation specialists, demo owners, sales teams, marketers, business stakeholders, and technical subject-matter experts.
- Gathered highly technical requirements and converted them into functional visual, interaction, and communication outputs.
- Supported software and demonstration handoffs through documentation, asset organization, workflow clarification, and cross-functional communication.
- Combined engineering understanding and design judgment to bridge physical machinery, automation logic, software behaviour, and human interpretation.

GRADUATE ENGINEER TRAINEE | Industrial Automation & Digital Technology, India

08/2024 – 01/2025

- Worked on digital experiences and visual communication for industrial automation, manufacturing software, smart manufacturing, and enterprise technology.
- Created interface screens, dashboards, user-facing layouts, visual explainers, technical graphics, presentation systems, and motion content.
- Used Figma, Adobe Creative Suite, Blender, frontend technologies, and presentation tools to translate engineering information into accessible experiences.
- Developed 3D visualizations and animations to explain equipment, processes, industrial environments, and technical concepts.
- Supported browser-based and interactive concepts combining software interfaces, visualization, technical information, and user interaction.
- Worked with engineering and business stakeholders to understand technical workflows and transform them into coherent user-facing narratives.
- Supported demo development across industrial software, production systems, automation workflows, digital transformation, and visualization use cases.
- Produced multimedia content for technical communication, product demonstrations, internal communication, webinars, and marketing-support initiatives.
- Applied graphic-design principles including layout, hierarchy, typography, composition, and information organization to technical material.
- Developed experience operating between engineering, design, communication, and digital production rather than within a single conventional discipline.

ENGINEERING INTERN | Industrial Automation & Digital Technology, India

01/2024 – 08/2024

- Supported industrial automation and software demonstrations through technical documentation, visual communication, interface assets, presentations, and digital content.
- Created graphic and 3D assets to simplify equipment behaviour, process workflows, automation concepts, and industrial software.
- Applied an Electronics Engineering foundation to understand technical systems while developing communication for non-specialist audiences.
- Created interface visuals and presentation assets connecting industrial technology with user-facing communication.
- Worked with technical teams and interpreted engineering information for visual output.
- Built an early foundation in combining engineering knowledge, design, visualization, and communication.

CREATIVE ASSOCIATE | Digital Media & Creative Production, India

08/2023 – 12/2023

- Created digital graphics, layouts, visual concepts, captions, and audience-facing communication assets.
- Worked across visual storytelling, digital content, platform-native design, and brand communication.
- Applied typography, hierarchy, composition, image treatment, and graphic-design principles across content.
- Supported creative ideation and execution for digital communication.
- Developed experience adapting visual language to audience, medium, and communication objective.
- Strengthened the connection between graphic design, storytelling, multimedia production, and audience perception.

ENGINEERING SIMULATION INTERN | Engineering Research & Simulation, India

Project-Based Experience

- Used MATLAB to develop engineering projects centered on thermodynamic systems and mathematical modeling.
- Modeled and analyzed Carnot-engine behaviour using thermodynamic relationships and computational methods.
- Developed MATLAB-based representations of quasistatic thermodynamic processes to explore changes in physical state variables.
- Applied mathematical equations and engineering principles to solve thermodynamics problems computationally.
- Translated theoretical concepts including heat, work, efficiency, pressure, volume, and temperature into programmable models.
- Used simulation and computational visualization to investigate idealized physical processes and engineering behaviour.
- Strengthened experience in physics-based modeling, mathematical reasoning, engineering computation, and simulation-driven problem solving.

CREATIVE ASSOCIATE | Graphic Design Studio, India

08/2020 – 12/2020

- Created posters, promotional graphics, layouts, and visual communication assets for client projects.
- Used Adobe Illustrator to create and refine vector-based graphics, typography, compositions, and production-ready design assets.
- Developed foundational proficiency in Adobe Illustrator, vector design, poster composition, layout, and visual hierarchy.
- Interpreted client requirements and translated them into visual concepts appropriate to different brands and communication objectives.
- Applied graphic-design fundamentals including typography, balance, hierarchy, composition, shape, colour, and image treatment.
- Worked within a small design-studio environment, gaining early experience in client-facing commercial creative production.

COMMERCIAL VOICE-OVER ARTIST | Audio Production Studio, India

2007 – 2015

- Recorded commercial voice-over spots for advertising and branded communication across multiple client campaigns.
- Performed studio voice work for established brands across consumer goods, retail, real estate, technology, lifestyle, and other commercial sectors.
- Adapted vocal tone, pacing, emphasis, energy, and delivery style according to script, brand identity, audience, and campaign requirements.
- Worked under studio direction to deliver concise, repeatable, production-ready commercial reads.

- Developed professional experience in spoken communication, script interpretation, vocal performance, timing, and audience perception.
- Contributed voice performances to advertising campaigns for nationally recognized brands through a professional recording-studio environment.
- Built an early foundation in audio communication and performance that later informed work in voice-over scripting, video production, storytelling, and multimedia communication.

SELECTED PROJECTS

- **Industrial Machine Vision Label Assurance System | Machine Vision / Electronics / Industrial Automation / UX** – Worked with an industrial machine-vision setup using Cognex In-Sight hardware and inspection software for automated label verification. Configured and tuned OCR and inspection behaviour, restored camera communication, validated pass/fail logic, corrected OPC mappings, and connected machine-generated inspection results to operator-facing industrial software. The project combined camera systems, inspection logic, automation communication, industrial software, troubleshooting, quality assurance, and human-readable feedback.
- **Cognex Vision System Integration | Machine Vision / Industrial Systems** – Worked with Cognex In-Sight 7402 and IS2000 vision environments, camera communication, OCR inspection, inspection parameters, CountPassFail data, and OPC-based communication. Investigated and corrected integration issues between camera-generated inspection information and downstream industrial applications.
- **Interactive Dairy Plant Visualizer | 3D Web / Industrial UX / Frontend** – Developed an interactive browser-based plant visualization for an industrial demonstration. Users could navigate plant areas through interactive points of interest and launch contextual process media. The experience combined spatial interface design, frontend development, multimedia integration, industrial workflow understanding, interaction logic, and operator-oriented information presentation.
- **Spray Dryer Digital Twin Visualization | 3D / Digital Twin / Process Visualization** – Created animated 3D visualization for a dairy spray-dryer digital-twin demonstration. Developed multiple process scenarios representing changes in spray density, droplet behaviour, powder behaviour, airflow, and operational conditions. Designed the visuals to reveal internal equipment behaviour and make process phenomena that are normally difficult to observe directly understandable to viewers.
- **Spray Dryer Interactive Process Assets | SVG Animation / Industrial Visualization** – Developed animated visual assets representing nozzle spray and droplet behaviour for integration into an industrial interface. Worked with scalable vector graphics, animation timing, opacity, movement, transparency, and density changes to create reusable process visualizations for technical demonstrations.
- **Spray Dryer KPI Dashboard Concepts | UX/UI / Data Visualization / Digital Twin** – Developed dashboard concepts for a digital-twin demonstration focused on spray-dryer performance. Explored production, quality, efficiency, process-condition, equipment-state, and optimization KPIs and structured them into operator-readable information hierarchies.
- **Integrated Operations Centre Usability Evaluation | UX Research / Enterprise Software / Industrial Operations** – Conducted usability-oriented evaluation of multi-user industrial operations workflows across virtualized environments. Documented workflow failures, access dependencies, reliability problems, performance issues, application behaviour, and their practical impact on users navigating the system.
- **Line Management Software Walkthrough | Enterprise UX / Industrial Software / Technical Communication** – Developed structured communication for an industrial line-management software environment covering line visualization, production dashboards, shifts, orders, process-cell status, dynamic lines, downtime reporting, timeline analysis, measured-value analysis, documentation, and simulation controls. Translated complex software functionality into an ordered, comprehensible product narrative.
- **Production Dashboard Communication System | Information Design / Industrial UX** – Structured technical explanations and visual communication around production dashboards, process states, running and stopped lines, downtime conditions, production orders, shifts, and operational performance. Focused on helping users interpret complex production information rapidly.
- **Electronic Batch Manufacturing Record Communication | Life Sciences / Motion / Technical Storytelling** – Developed a multi-stage video narrative around electronic batch manufacturing records for life-sciences manufacturing. Structured communication around paper-based manufacturing challenges, digital workflows, ISA-88 batch processes, production review, QA review, electronic signatures, audit trails, deviations, user authentication, review by exception, and digital batch reporting.
- **AI-Assisted Technical Video Workflow | Generative Media / Storyboarding / Industrial Communication** – Designed shot structures, visual prompts, voice-over mapping, on-screen messaging, and generated-video requirements for technical communication. Balanced source assets, generated footage, software visuals, engineering accuracy, visual continuity, and narrative structure.
- **Industrial Process Optimization Video | Industrial Process / Video / Technical Storytelling** – Developed communication and production planning for technical content around industrial process optimization, combining engineering concepts, software capabilities, visual narrative, and product communication.
- **Conversational 3D Avatar | Human-Computer Interaction / Three.js / Creative Technology** – Built an interactive browser-based 3D avatar using JavaScript, Three.js, GLTF assets, animation states, voice and text interaction, persona selection, user controls, and real-time feedback. Combined frontend engineering, character interaction, 3D rendering, UX, interface states, and creative technology.
- **Interactive Globe Portfolio | React / Three.js / Spatial Interaction** – Designed and developed an interactive 3D globe as a portfolio navigation system using React and Three.js. Implemented interactive geographic markers, routed project navigation, spatial

interaction, camera behaviour, browser rendering, and frontend interface logic.

- **Interactive 3D Product Viewer | Product Visualization / Three.js / Frontend** – Developed an interactive browser-based 3D product viewer using Three.js and Vite. Integrated a GLTF model, environment-based presentation, orbit controls, camera behaviour, lighting, interaction, and browser deployment to create an exploratory digital product experience.
- **Browser-Based 3D Product Experiences | Creative Development / Product UX** – Explored real-time 3D rendering as an interface medium, combining direct manipulation, camera logic, spatial interaction, responsive UI, product visualization, and frontend development.
- **Industrial Digital Twin Demonstrations | Automation / 3D / UX** – Created and supported demonstrations where 3D visualization, industrial software, process information, equipment behaviour, and user interfaces work together to communicate the state of physical manufacturing processes.
- **Dashboard and KPI Interface Concepts | UX/UI / Information Visualization** – Designed dashboard concepts for industrial systems using process, production, quality, equipment, efficiency, and operational information. Applied hierarchy, visual encoding, grouping, prioritization, and operator-focused layout principles to dense technical data.
- **Industrial HMI and Interface Visuals | UI Design / Industrial Software** – Created UI-style screens, dashboard visuals, interface layouts, and supporting graphics for industrial and enterprise software demonstrations, balancing technical fidelity with readability and visual communication.
- **Carnot Engine Simulation | MATLAB / Thermodynamics / Engineering Simulation** – Developed a MATLAB-based engineering project modeling the behaviour of an idealized Carnot heat engine. Applied thermodynamic relationships and mathematical computation to investigate process states, heat transfer, work, efficiency, and the interaction between variables across the thermodynamic cycle.
- **Quasistatic Thermodynamic Process Modeling | MATLAB / Physics-Based Modeling** – Used MATLAB to model and solve problems involving quasistatic thermodynamic processes. Translated physical laws and mathematical relationships into computational representations to analyze process behaviour and changes in thermodynamic state variables.
- **Engineering Simulation with MATLAB | Computational Engineering / Mathematical Modeling** – Applied MATLAB as an engineering computation environment to convert theoretical thermodynamics problems into programmable models, calculations, and visualizable engineering processes. Combined mathematical reasoning, physical-system understanding, and computational problem solving.
- **Technical Explainer and Product Videos | Motion / Video / Product Communication** – Produced and supported product explainers, software walkthroughs, industrial demo videos, promotional content, webinar media, and technical narratives using video editing, motion graphics, voice-over scripting, screen recordings, generated footage, presentation assets, and technical graphics.
- **Technical Storyboarding System | Content Strategy / Motion** – Developed storyboards and shot-level production structures mapping technical concepts to visual sequences, voice-over, text overlays, demonstrations, animations, and generated media. Used storyboarding as a bridge between subject-matter expertise and final communication.
- **Technical Presentation Systems | Graphic Design / Information Design** – Created presentation layouts, technical diagrams, visual explainers, process graphics, product communication assets, and reusable visual systems for engineering, sales, marketing, internal communication, and webinar use.
- **Industrial Process Visual Storytelling | Engineering Communication / Visual Design** – Translated processes that are abstract, internal, data-driven, or physically difficult to observe into understandable visual sequences using diagrams, UI, animation, 3D, dashboards, text, and motion.
- **Client Poster and Graphic Design | Graphic Design / Adobe Illustrator** – Created client-facing posters, promotional graphics, vector assets, layouts, and visual compositions in a commercial graphic-design environment. Developed foundational Adobe Illustrator proficiency while learning to interpret client requirements and execute graphics within different visual identities and communication objectives.
- **Commercial Voice Performance | Voice-Over / Audio Communication** – Recorded commercial advertising spots across consumer goods, retail, real estate, technology, lifestyle, and other sectors. Developed experience in professional studio recording, directed performance, script interpretation, pacing, tonal adaptation, vocal clarity, repeatability, and concise brand-oriented communication.
- **Portfolio Web Development | Frontend / Interaction Design / Personal Brand** – Designed and developed a browser-based portfolio environment combining conventional frontend layouts with interactive 3D experiences. Worked across desktop and responsive-mobile behaviour, project navigation, visual hierarchy, interaction, and deployment.
- **Responsive Interface Development | Frontend / UX** – Tested and refined web layouts across desktop and mobile viewport sizes, identifying alignment, positioning, responsiveness, and interaction issues using browser developer tools and iterative frontend changes.
- **YAML-to-PDF Resume Generation System | Python / Automation / Information Design** – Built a reusable resume-generation workflow in Python using YAML as the content source and LaTeX as the presentation layer. Implemented structured rendering for skills, experience, projects, and education, automated XeLaTeX compilation, special-character escaping, and reusable content templating.
- **Content-Driven Portfolio Architecture | Information Architecture / Personal Brand** – Structured a portfolio around overlapping domains including electronics, engineering simulation, perception systems, machine vision, industrial technology, 3D, frontend development, interaction design, graphic design, motion, voice, and technical communication, using project-based evidence to demonstrate multidisciplinary capability.

ENGINEERING BACKGROUND

• Bachelor of Engineering | Electronics Engineering | Engineering Institution, India

Graduated: 2024